

Do humans have an
internal drive to share
what is on our minds?
Testing
Mitteilungsbedürfnis

Sabine Blütgen, Simon Kirby, Jennifer Culbertson
Centre for Language Evolution, University of Edinburgh
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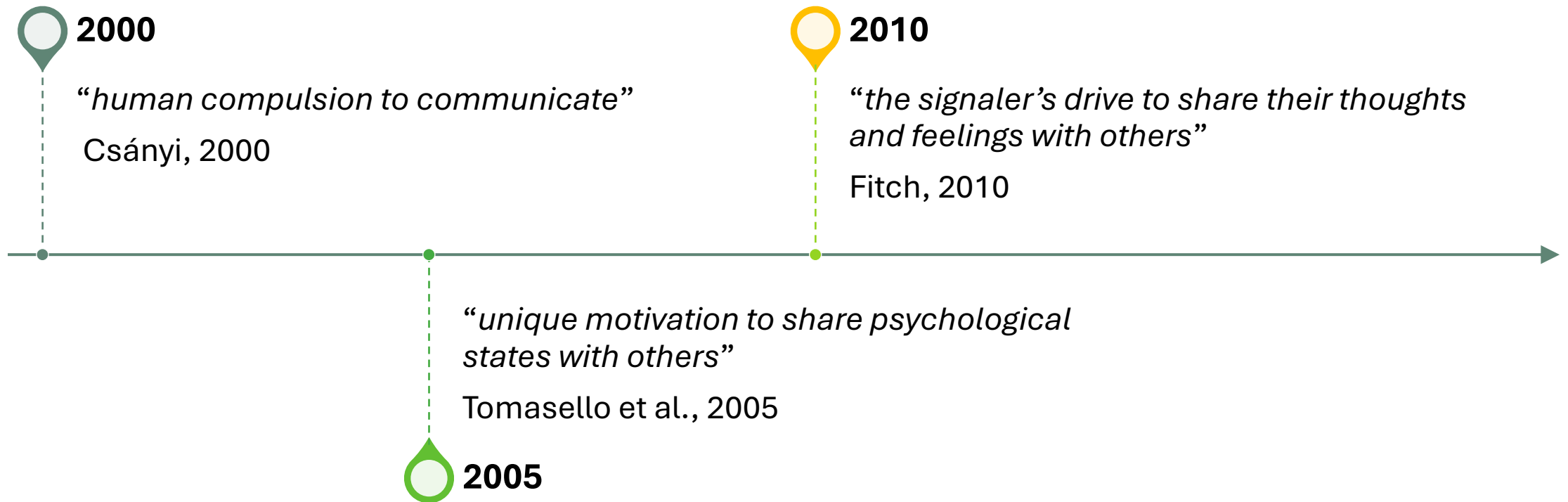
Raise your hand if...

You have shared your
thoughts and feelings
today

You have ever felt like
you needed to tell
someone something

You have ever
wondered “why did I
share that?”







Fitch, 2010

“the signaler’s drive to share their thoughts and feelings with others”



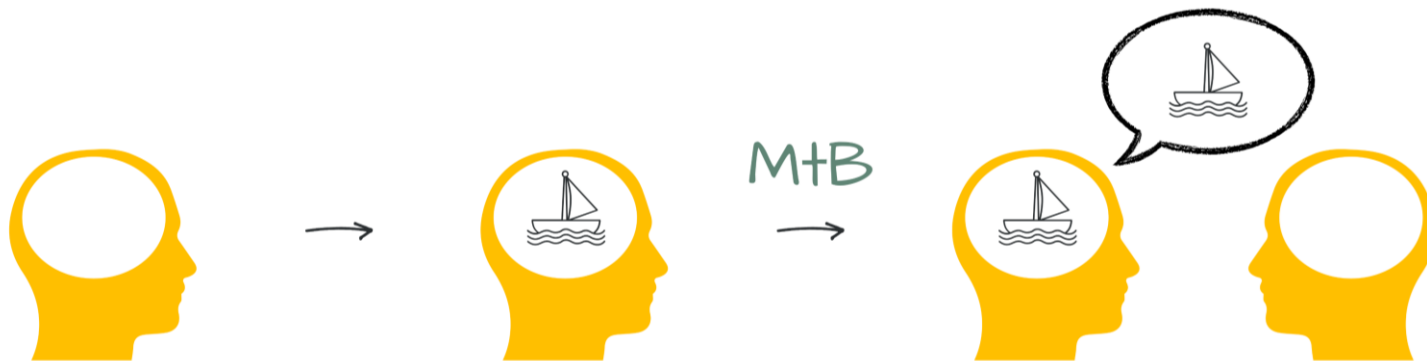
Mitteilungsbedürfnis (MtB)

German for *“the need to communicate/express oneself/share one’s thoughts and feelings”*



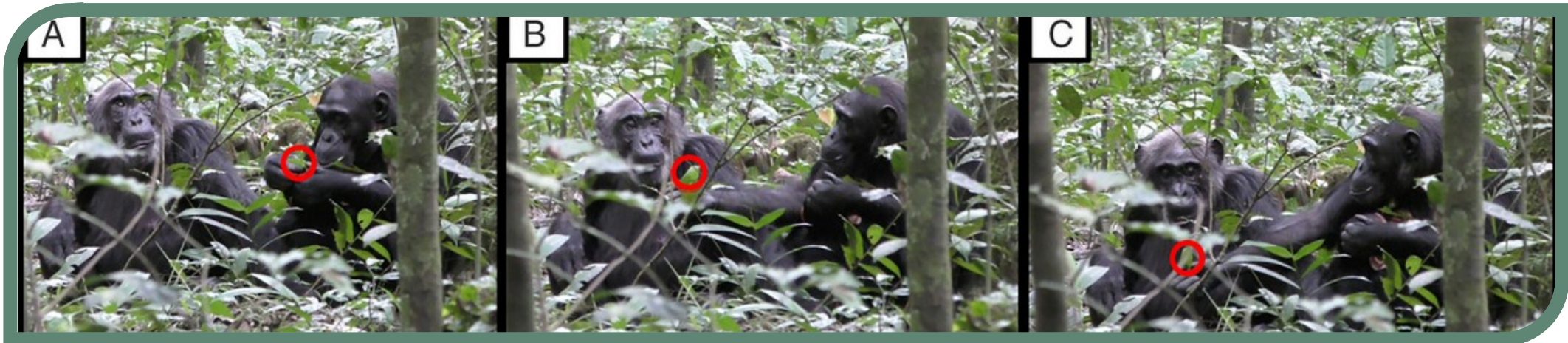
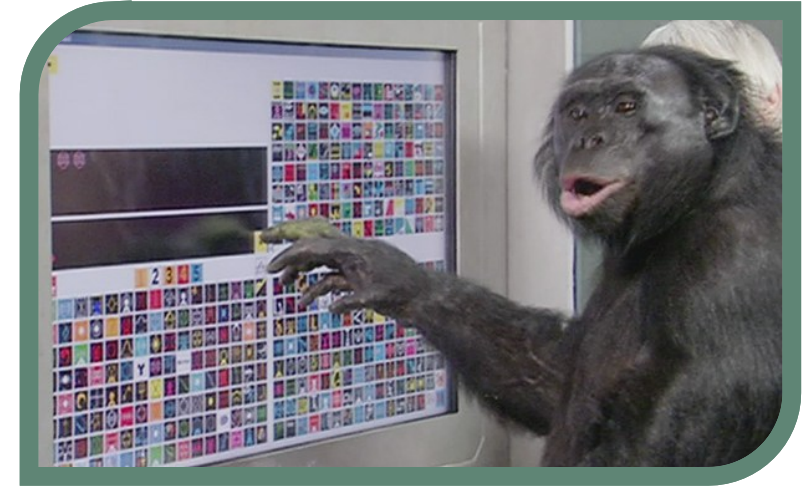
Our definition

intrinsic drive to share what is on our minds with others, which is *independent of external motivations*



Non-human apes

- Next to no recorded incidents of declarative sharing in the wild (exception: Wilke et al., 2022)
- Utterances of language-trained apes with caregivers overwhelmingly imperative (e.g. 96% of Kanzi's utterances; Greenfield & Savage-Rumbaugh, 1990)





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Human children



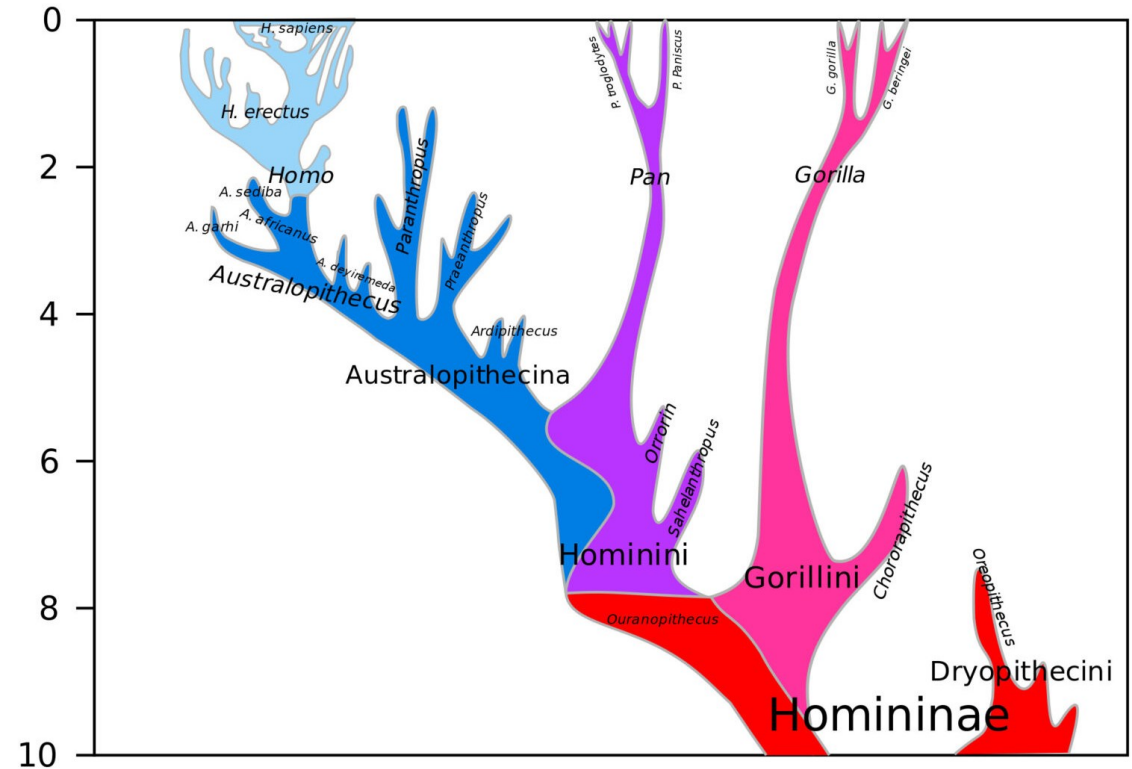
- declarative communication arises very early, potentially with proto-conversations (from few months after birth)
- more obviously with pointing and hold-out gestures from around 6 months
- allows for collaborative engagement from 12 months onwards (Tomasello et al. 2005)

○



Evolutionary explanations

- Socioecological explanation for cognitive and behavioural differences in human and other apes (Csányi, 2000; Dor 2023; Tomasello et al. 2012)
 - At some point after split of the *Pan* and *Homo* lines (~6 mya), being able to collaborate with others became obligatory for archaic humans
 - Strong social selection for individuals with cooperative skills and motivations



Existing empirical evidence

Behavioural/Neuroimaging evidence (Tamir & Mitchell, 2012; Tamir et al., 2015)

- Participants chose to **forgo money** to answer questions about themselves or inform others
- Brain regions previously shown to be associated with **reward** activated when participants shared
- Participants shared information that was **random** and did **not provide an obvious benefit** to them or the receiver

Existing empirical evidence

Role of partner choice in information sharing (Jimenez et al. 2025 [preprint])

- Series of experiments to test if knowledge holders would share with those who had high reputations for sharing and withhold information from those who had low reputation
- Participants **shared their knowledge quite indiscriminately**, even when they could have obtained higher pay-offs by being selfish



Existing empirical evidence – limitations

External motivations

- Jimenez et al. 2025 – clear motivation to help others
- Tamir et al. studies – strong social component

Our study



Paradigm



Experiment task

On the next screen, you will see an image. Please have a look at it - you can take as much time as you need. You will then be asked to name something you have discovered in the image.

Continue



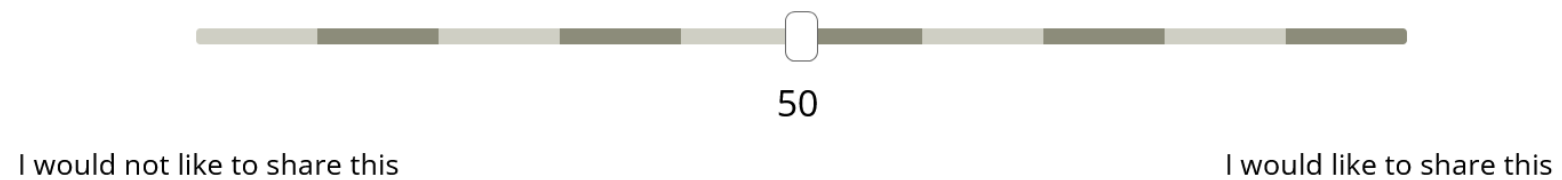


Please name one thing you have discovered:

blue flower

Continue

Would you like to share your discovery of "blue flower" with someone else?



Continue

We are interested in how and why people share information. Could you tell us why you would like to share this?

Continue

Experiment task

You will now see the image again, and we will again ask you to name something you have discovered in the image. You can choose the same item or something different.

Continue

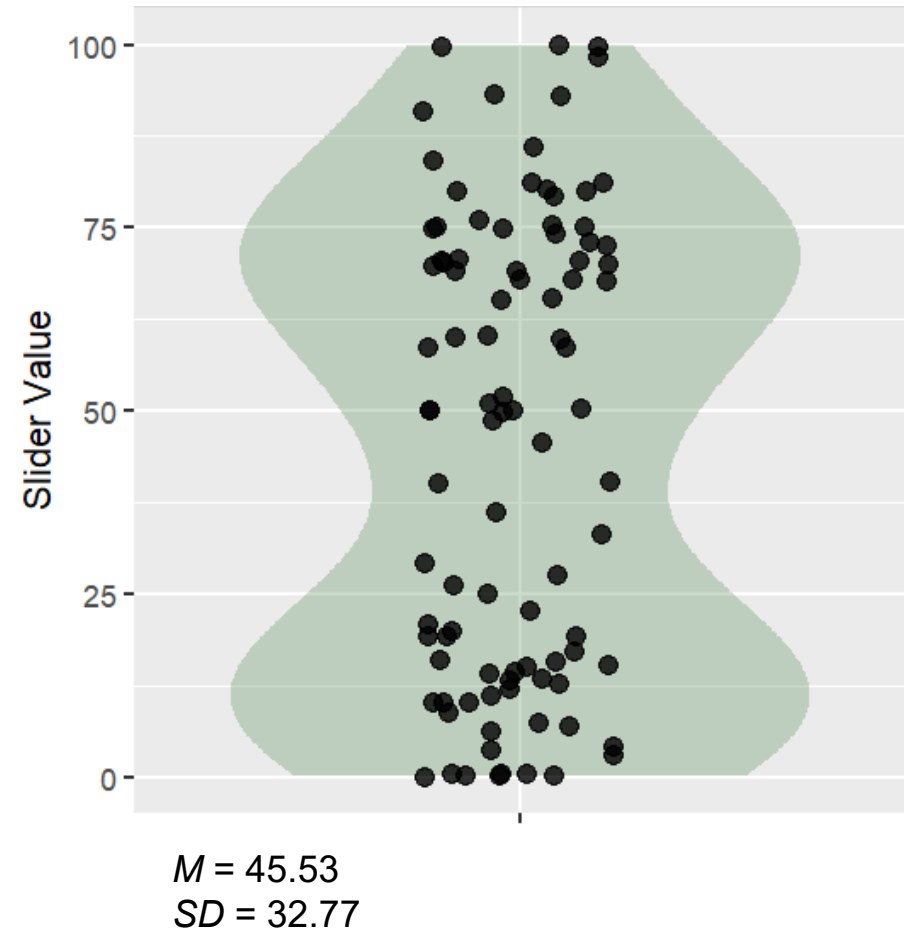
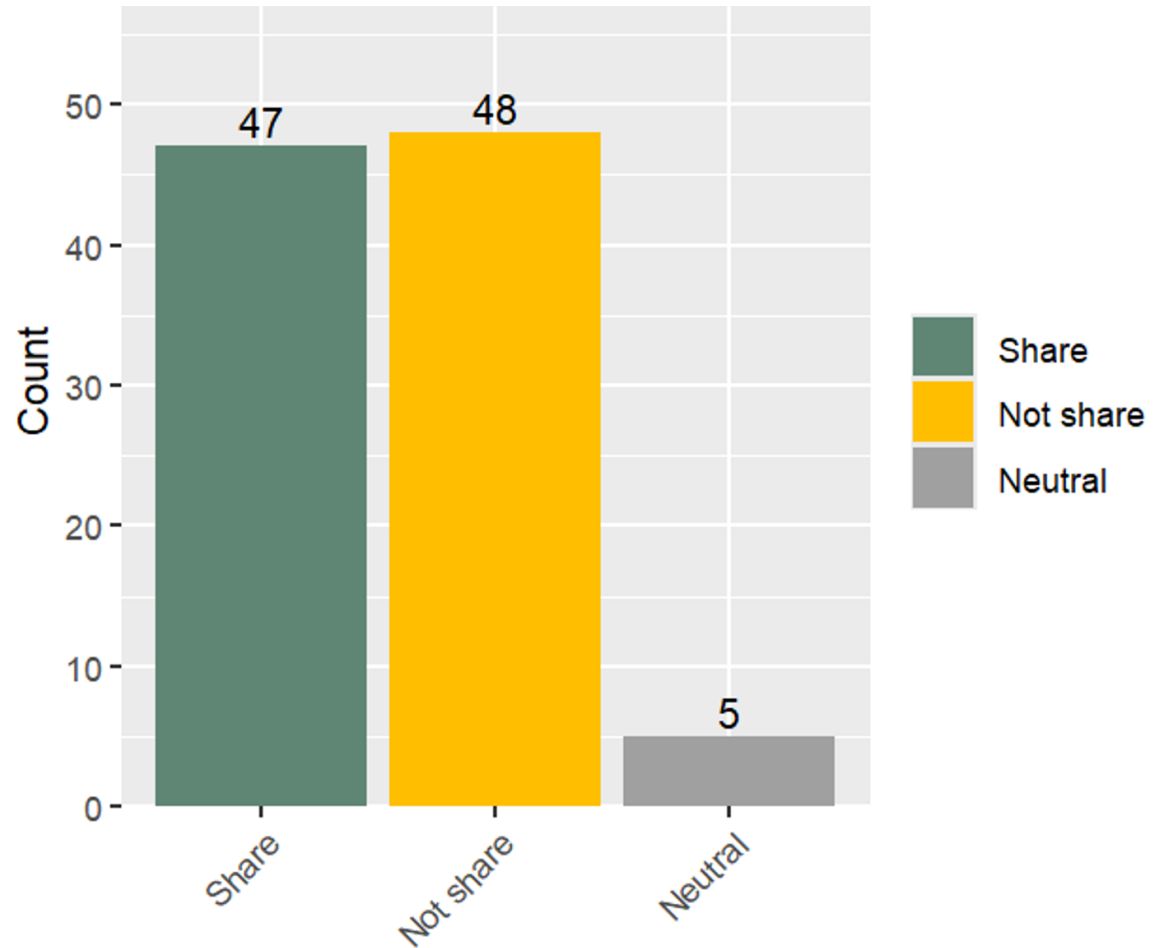
Participants

- 50 participants on Prolific from US, UK, and EU (self-reported)

Results

Will participants share information even without a clear external motivation?

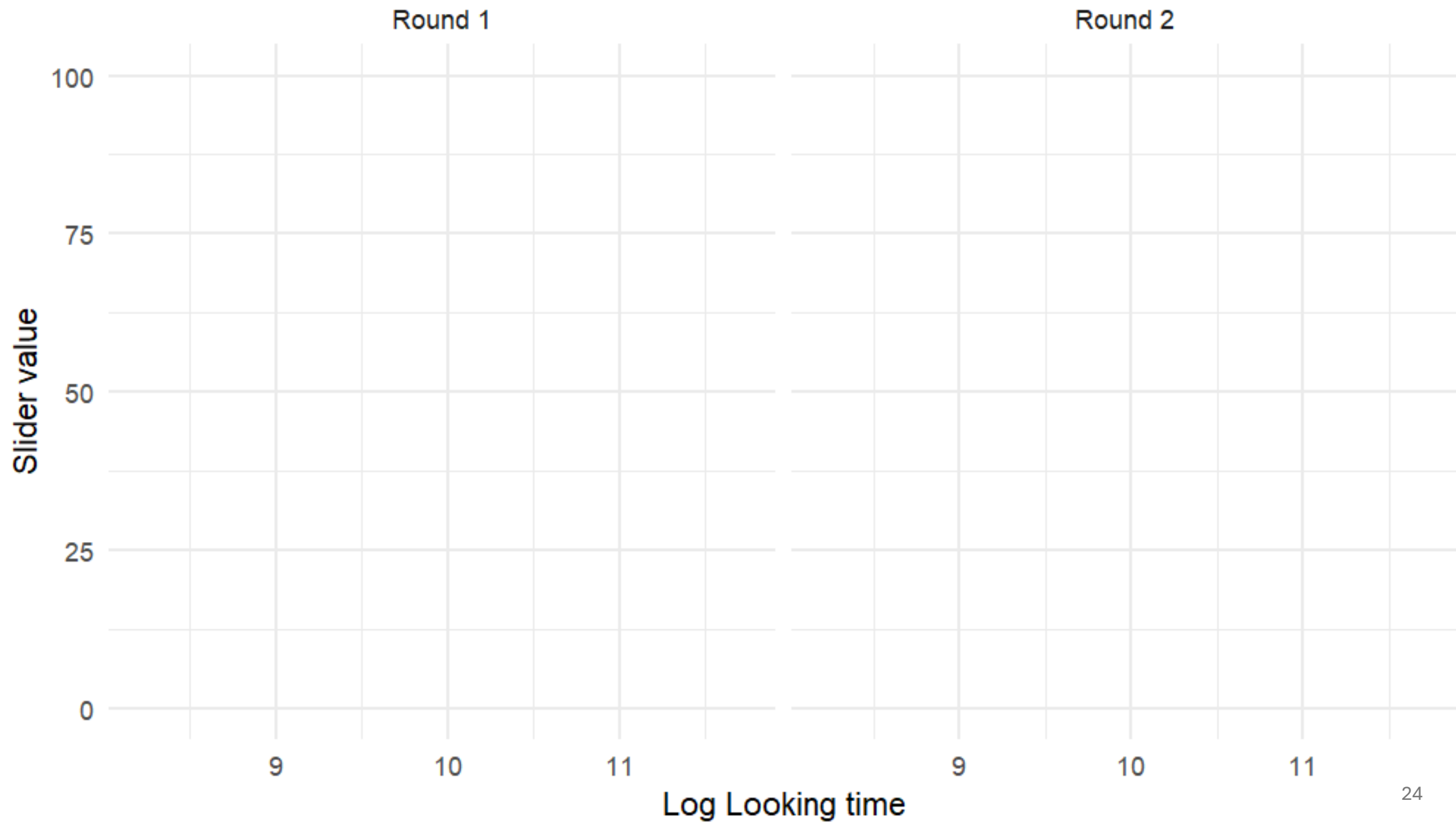
Results

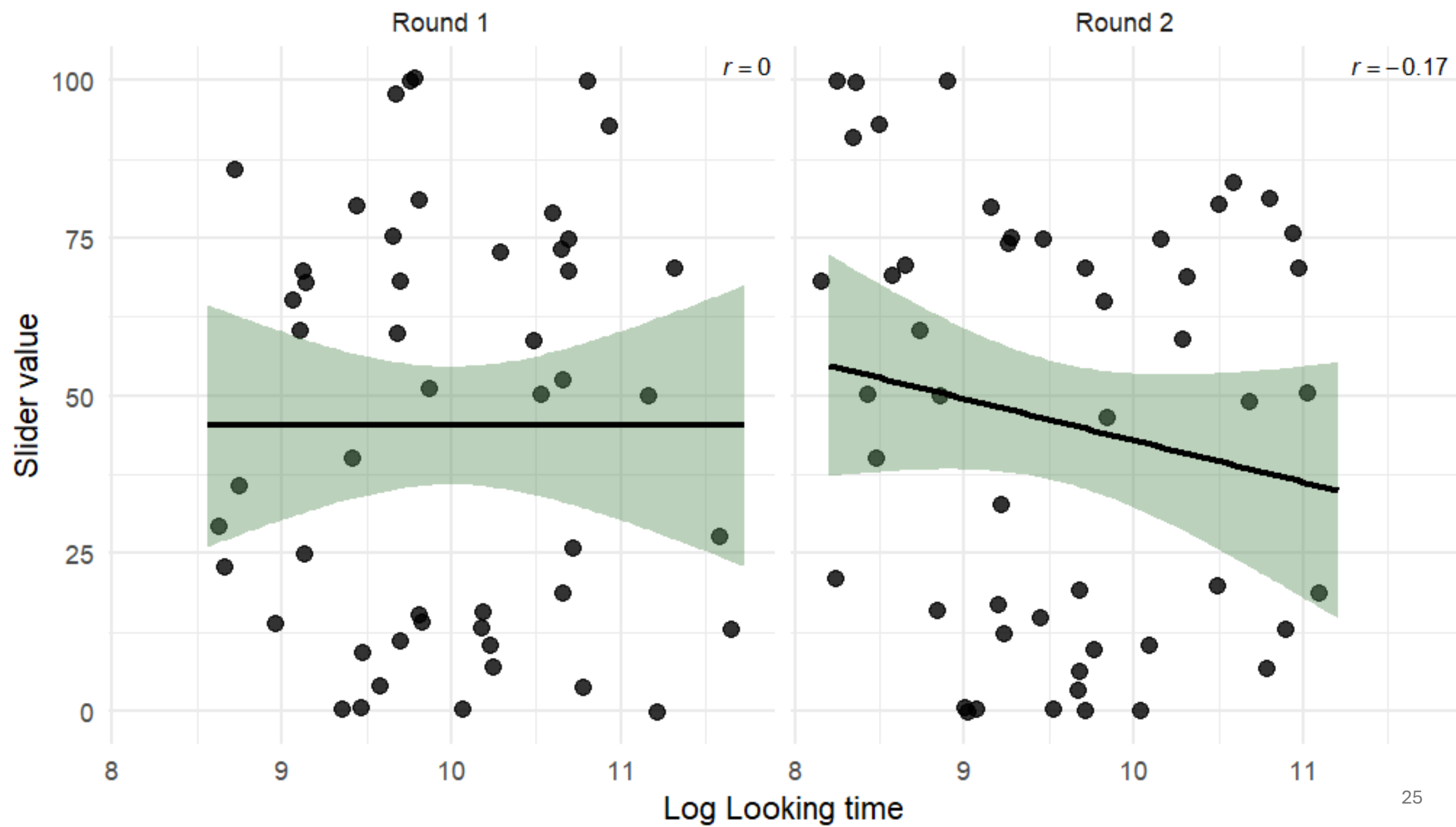




Collection effort

- Collecting information reveals what we pay attention to
- Sharing information that is perceived as valuable might advertise sharer as valuable themselves (Dessalles, 2010)
- Prediction: Longer looking time = higher slider value



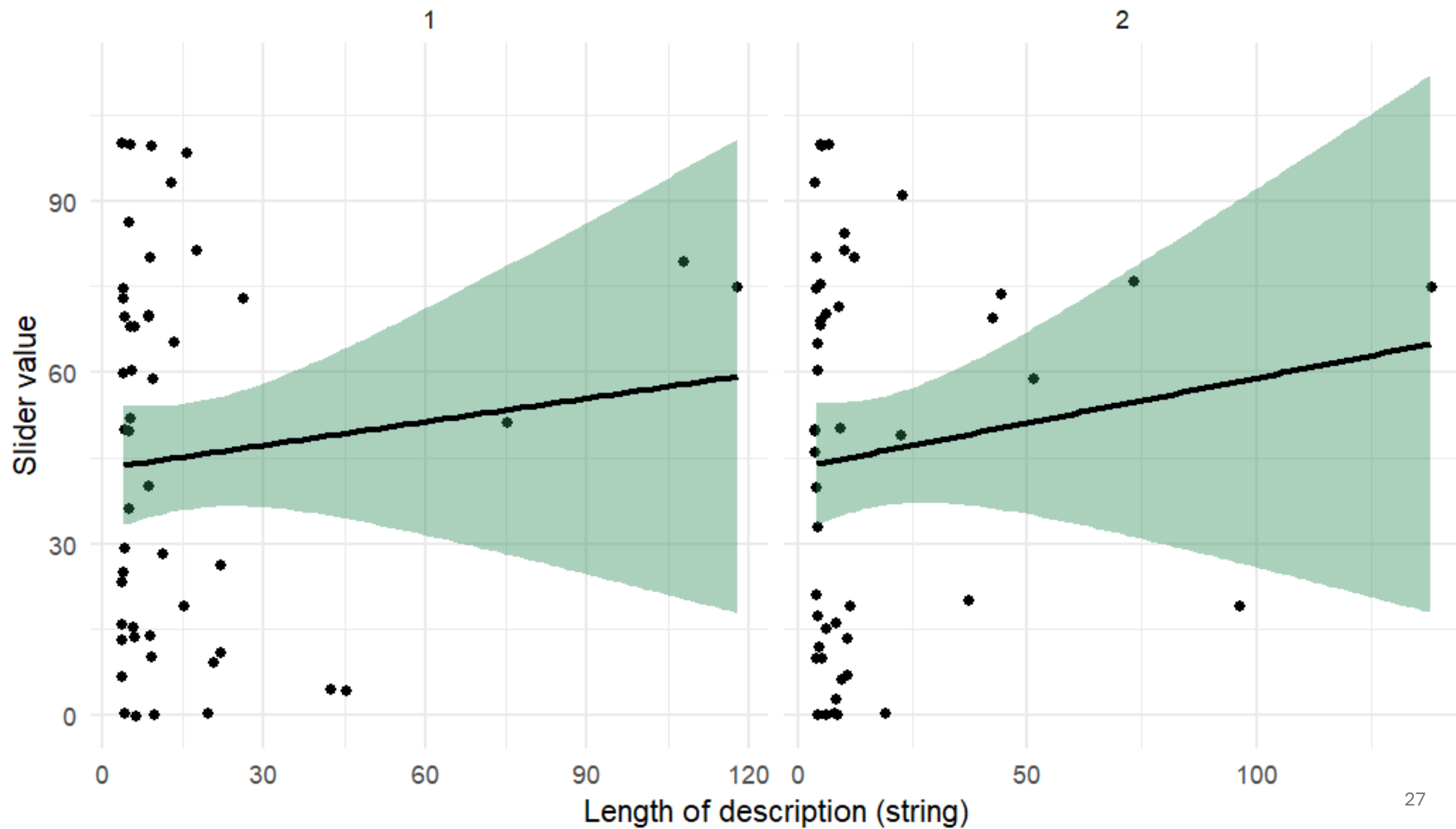


Length of description and sharing choice

Toad

Two birds on top standing on
branches that no artist
would ever put them on
because the branches seem
way too thin to support
those birds

Slider values and length of description by round



Remove information collection?

- Follow up experiment (50 participants on Prolific from US, UK, and EU (self-reported))
- Paradigm identical except for information collection
- Prediction: Participants will share less

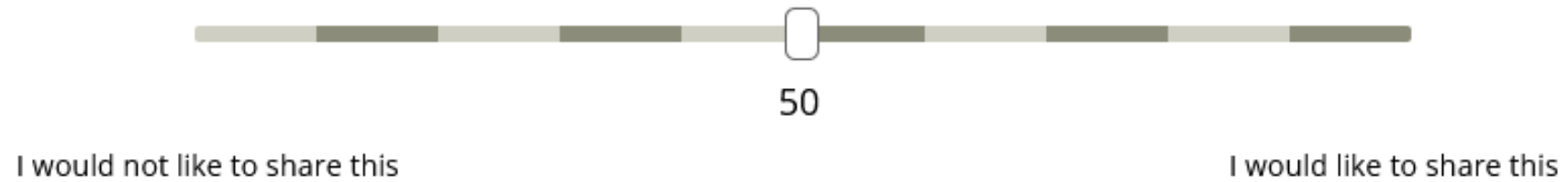


Here is one thing from the image:

frog

Continue

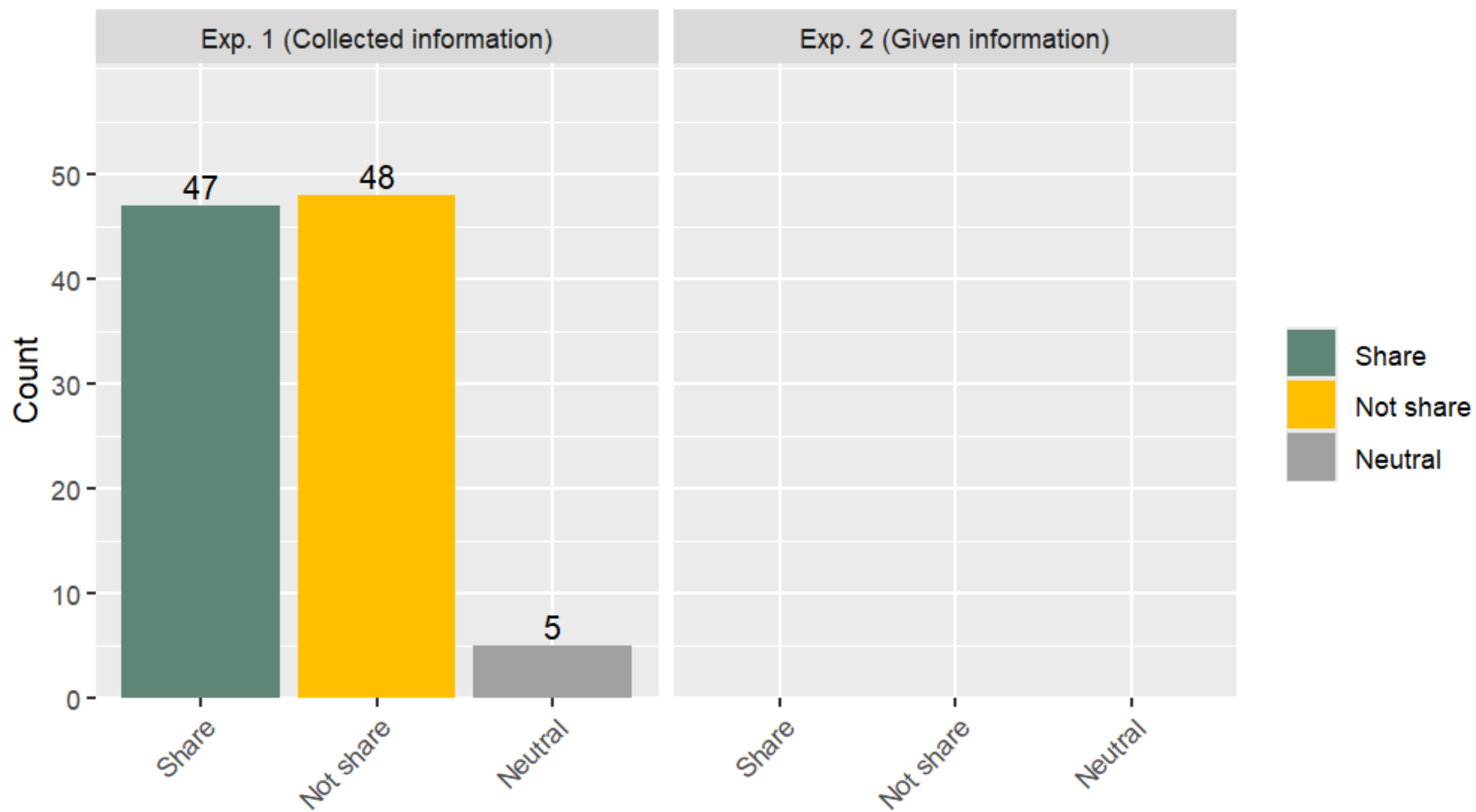
Would you like to share that there is a **frog** in the image with someone else?

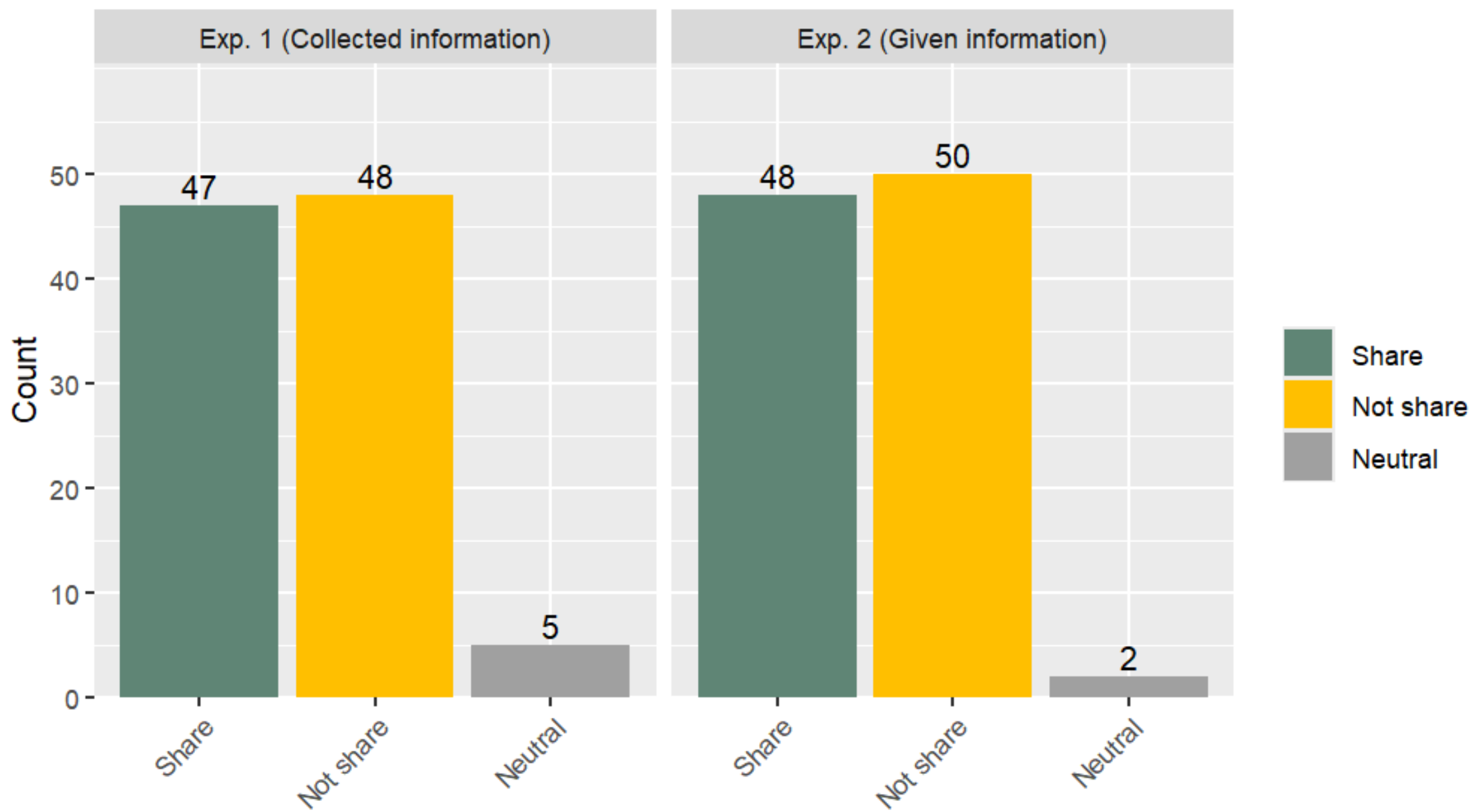


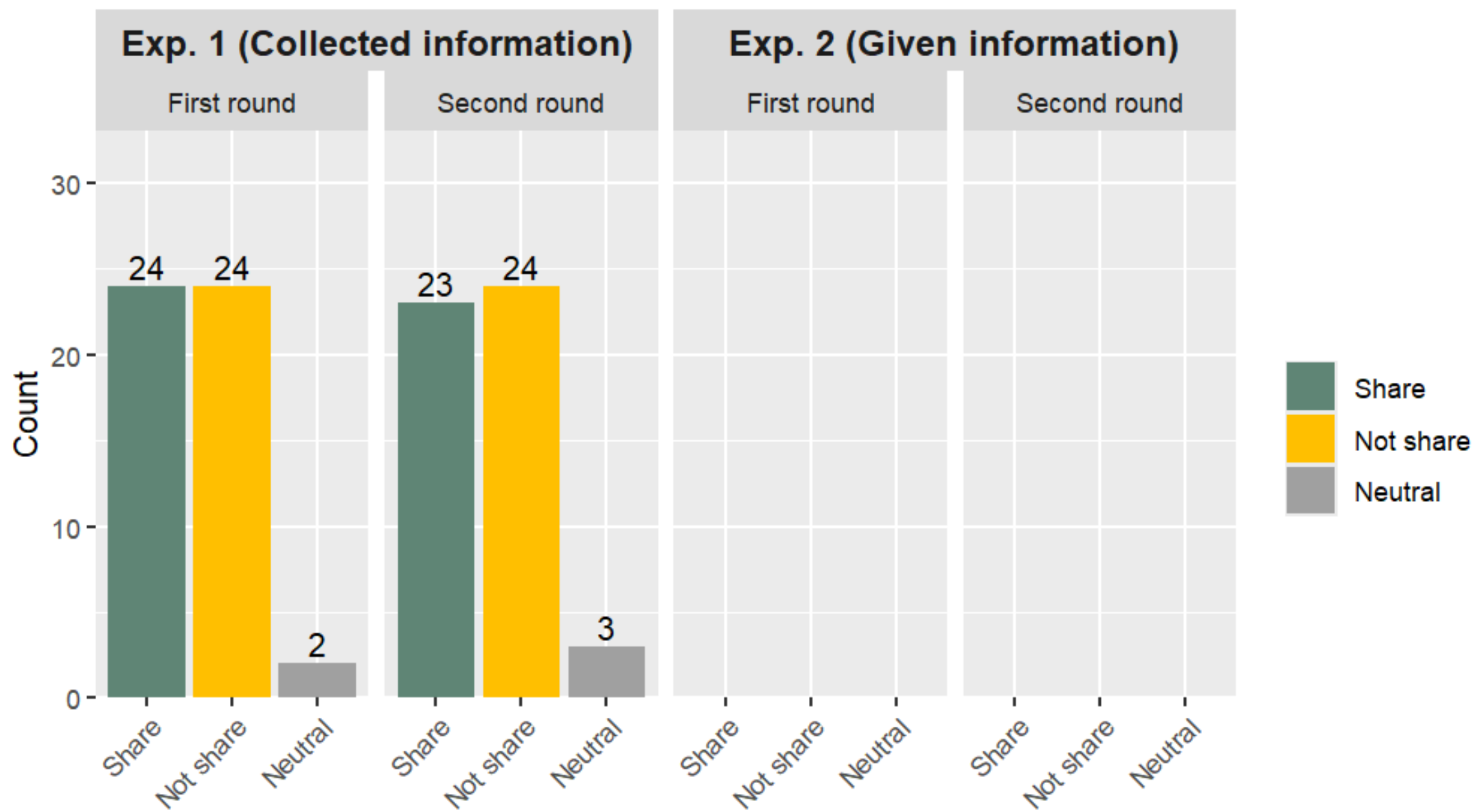
Continue

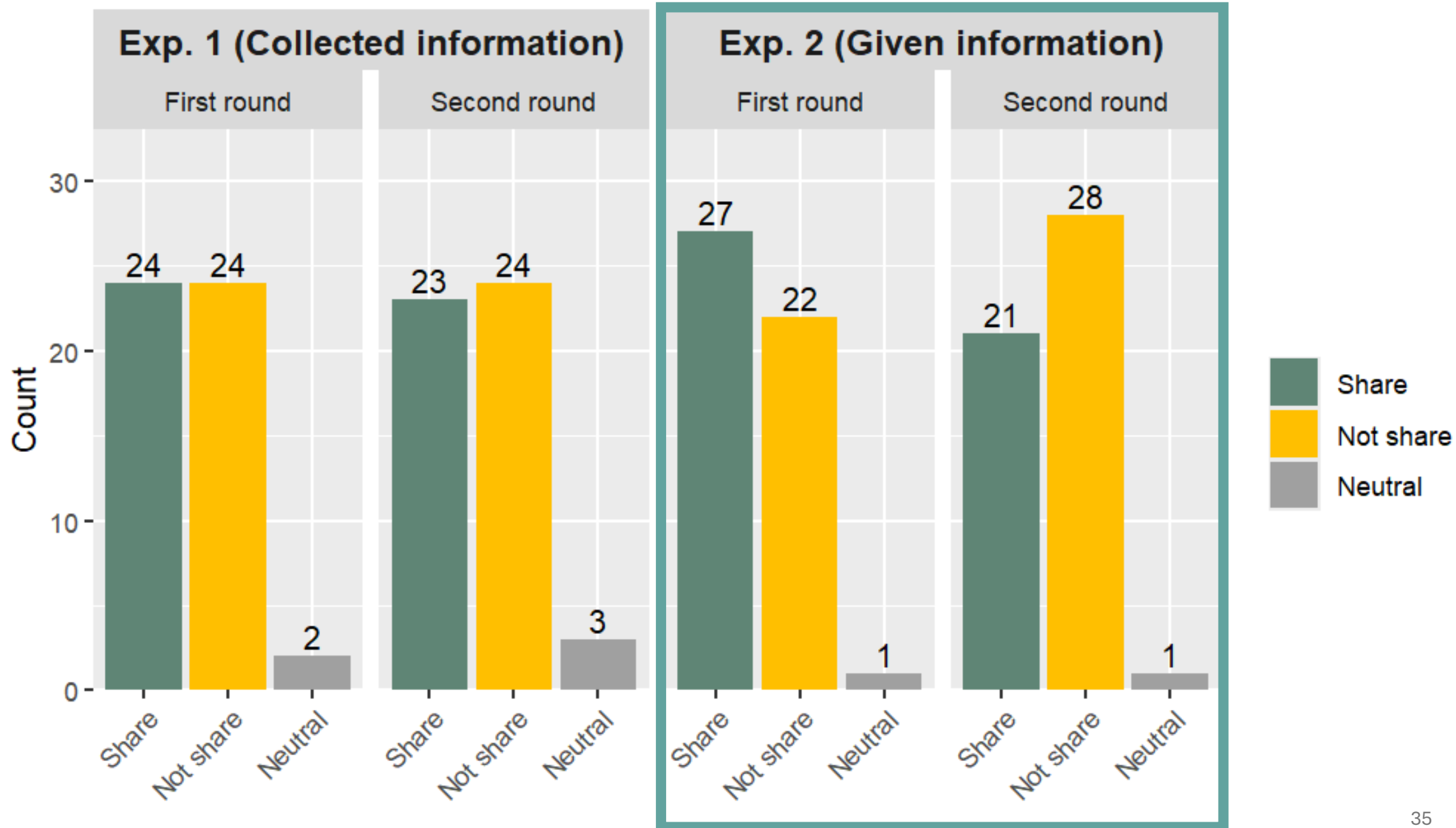
Results

Will collected
information be
shared more readily?

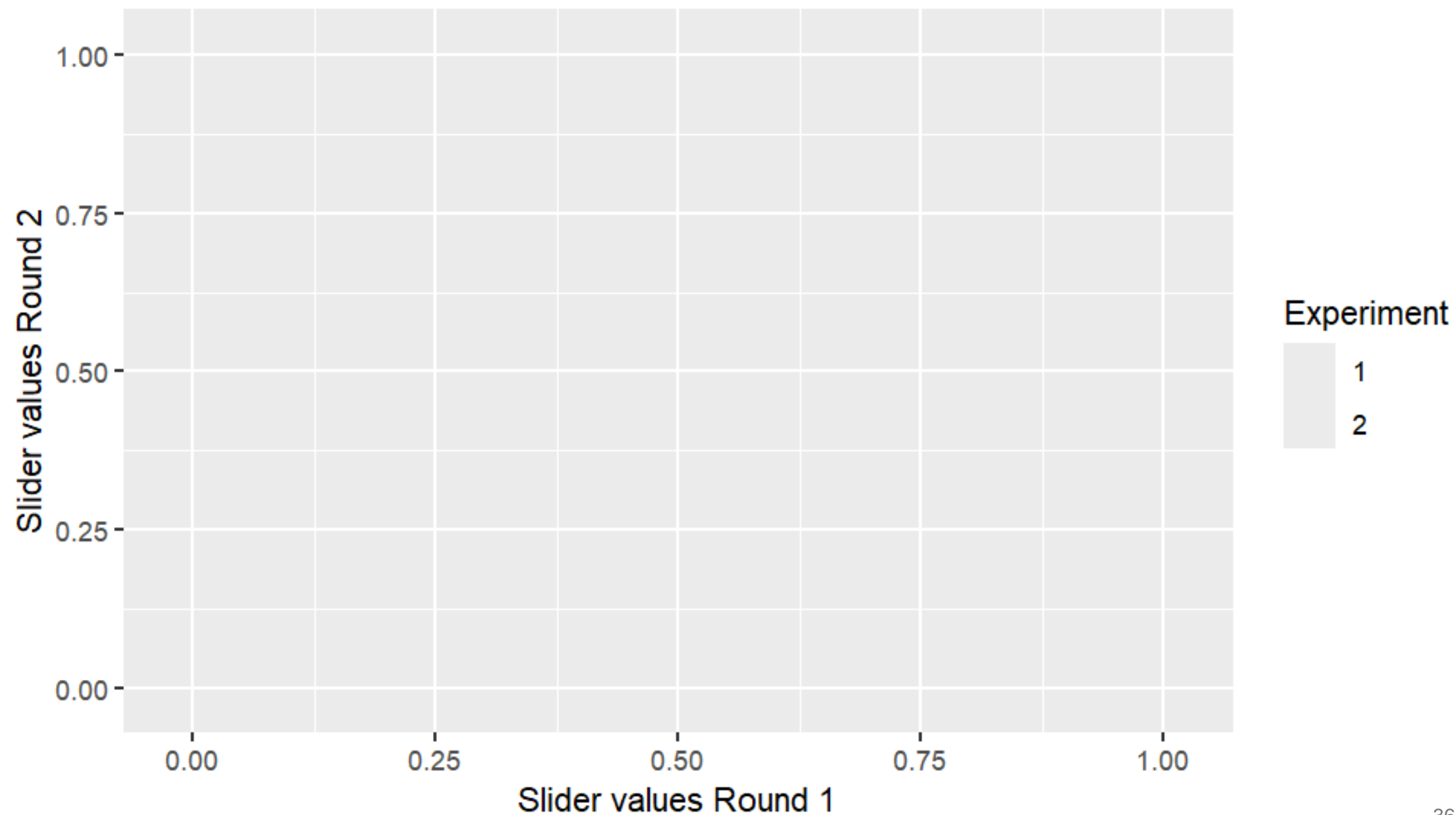








Slider value consistency by round (Model output)



Slider value consistency by round (Model output)



Why did participants choose to share/not to share?

A word cloud shaped like a speech bubble pointing downwards. The words are in various sizes and orientations. The most prominent words are 'help others', 'personal values', 'irrelevant', and 'share something positive'. Other words include 'not obvious', 'too obvious', 'certainty of knowledge', 'not interesting/special', 'interesting/unusual/unexpected', 'relevant', 'no one to share with', and 'encourage searching'.

not obvious
too obvious
certainty of knowledge
help others
personal values
not interesting/special
interesting/unusual/unexpected
share something positive
relevant
no one to share with
irrelevant
encourage searching

1

A word cloud shaped like a speech bubble pointing downwards. The words are in various sizes and orientations. The most prominent words are 'dislike object', 'accuracy of information', 'irrelevant', and 'share something positive'. Other words include 'like object', 'personal values', 'not interesting/special', 'interesting/unusual/unexpected', 'relevant', 'too obvious', 'not obvious', 'humour', 'something to talk about', 'dislike image', 'impression management', 'not useful', 'mtb', and 'help others'.

like object
dislike object
impression management
not useful
share something positive
personal values
humour
not interesting/special
something to talk about
accuracy of information
dislike image
interesting/unusual/unexpected
relevant
too obvious
irrelevant
not obvious
mtb
help others

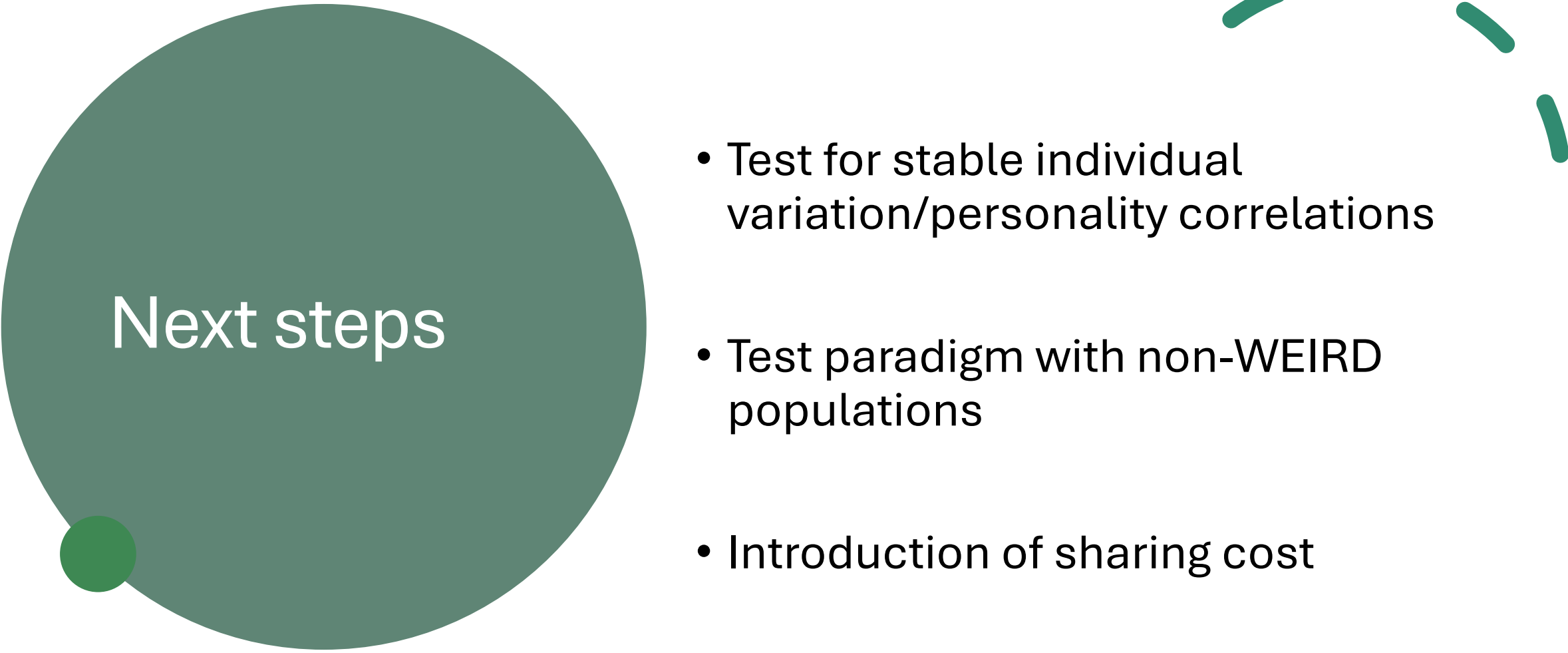
2

Summary & Discussion

Paradigm works

No clear support that collection effort influences sharing choice

Mixed picture for collected vs given information

A large dark green circle on the left side of the slide, with a smaller dark green circle at its bottom-left edge. In the top right corner, there are four green curved line segments arranged in a partial arc.

Next steps

- Test for stable individual variation/personality correlations
- Test paradigm with non-WEIRD populations
- Introduction of sharing cost

Thank you :)

s.bluetgen@sms.ed.ac.uk
sabinebluetgen.github.io/

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